

Top 20–25 Most Critical Semiconductor Barriers (Across All Categories)

1. High Chip Fabrication & Initial Setup Cost (*Resource Barriers*)

- Most fundamental barrier — fabs require billions in CAPEX.

2. Inadequate Semiconductor Manufacturing & Testing Infrastructure (*Resource Barriers*)

- India lacks advanced fabrication and testing capability.

3. Inconsistent Power and Water Supply (*Resource Barriers*)

- Chipmaking needs ultra-reliable utilities.

4. Talent Drain & Skill Gaps in Semiconductor Industry (*Resource Barriers*)

- A skilled workforce is critical and India faces major shortages.

5. Limited R&D and Innovation Ecosystem (*Resource Barriers*)

- No strong indigenous design-to-manufacture pipeline.

6. Slow Pace of Indian Semiconductor Mission (ISM & SPECS) (*Regulatory Compliance*)

- Delays in execution of national initiatives.

7. Complex Approval Processes & Limited Industrial Collaboration (*Regulatory Compliance*)

- Red tape hinders speed and coordination.

8. Stringent and Overlapping Regulatory Norms (*Regulatory Compliance*)

- Makes foreign/domestic investments harder.

9. Globally Fragmented Semiconductor Supply Chain (*Supply Chain & Ops*)

- Overdependence on other countries creates vulnerability.

10. Inadequate & Fragile Domestic Ecosystem (*Supply Chain & Ops*)

- India lacks a resilient chip manufacturing value chain.

11. Lack of Value Chain Integration (*Supply Chain & Ops*)

- No end-to-end integration — fragmented ecosystem.

12. Lack of Specific & Sophisticated Transportation Network (*Supply Chain & Ops*)

- Semiconductor transport needs special handling (temp, vibration, etc.).

13. Limited Access to Critical Raw Materials & Rare Earth Elements (*Supply Chain & Ops*)

- India is not self-sufficient in raw chip materials.

14. Geopolitical Rivalry & Semiconductor Dominance (*Global Conflicts*)

- US-China tensions affect global chip trade.

15. US-China Trade Conflicts & Restrictions (*Global Conflicts*)

- Export bans on advanced nodes and AI chips.

16. Taiwan Dominance & Alliances (*Global Conflicts*)

- Over 90% of advanced chips come from TSMC (Taiwan).

17. Global Currency Volatility & Pricing Instability (*Global Conflicts*)

- Affects global competitiveness and pricing.

18. IP Theft & Cybersecurity Threats (*Innovation Barriers*)

- Limits global trust and collaboration.

19. Limited Semiconductor Logic Design Capabilities (*Innovation Barriers*)

- Design is the brain of chips - India still lags here.

20. Lack of Circular Economy Models in Chip Manufacturing (*Innovation Barriers*)

- Waste and e-waste management issues.

21. Water Regulations & Recycling Challenges (SDGs) (*Innovation Barriers*)

- Chipmaking is water-intensive and unsustainable without recycling.

22. Sanctions on AI & Quantum Chip Technologies (*Global Conflicts*)

- Limits access to cutting-edge technologies.

23. Unclear Standards for IP Protection in Semiconductor Supply Chain (*Regulatory Compliance*)

- Deters foreign players from trusting Indian firms.

24. Techno-nationalism & Strategic Stockpiling (*Global Conflicts*)

- Countries hoarding chips as a strategic asset - reduces global cooperation.

25. Energy-Intensive Processes Clash with Net-Zero Goals (*Innovation Barriers*)

- Manufacturing is environmentally unsustainable unless green practices are adopted.

